

BETTY: The Super Power To Save Power



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Electricity bills form a major part of office and household running costs. For organisations, average electricity consumption accounts for 19 per cent of the running expenditure. Traditionally, in India meter readings are taken manually, which quite often end up as 'round-off' or 'approximate' figures. Moreover, users do not get to see exactly how they are spending on their electricity and thus figure out ways to lower the bills.

Thankfully, the smart age of connected devices has given us ample new options that inform us about our energy usage and ways to lower it. One such option is Betty by Beyond Evolution, which merits attention due to its affordable price.

The inception

Saurabh Sharma, co-founder and CEO of Beyond Evolution, saw the opportunity in controlling the power consumption of everyday appliances and easing the burden of electricity cost for the common man. Quite often users forget to switch off electrical appliances during power cuts, leaving the appliances running when the power resumes.

This not only wastes power but also poses fire risk.

"Our intent was not to become a so-called 'home-automation' solution provider. Betty has been designed for the more common challenges faced by majority of the people. Home automation may indicate a form of luxury, while Betty targets to address the day-to-day necessities," explains Sharma.

The whole thought process kick-started R&D initiatives, which took 18 months to develop the complete solution. Betty can *in-situ* measure appliance energy consumption, set auto-control (automatic 'on' and 'off') schedules as well as electricity usage thresholds, provide energy us-

age analytics, and, more interestingly, notify users of any security breach.

Being a newcomer in the manufacturing arena, the company had to scale a steep learning curve. Limited funds were another challenge. To start the fund inflow, the company kicked off with mobile application development and attracted significant customers. Once the production seemed financially feasible, they started off with the Betty project. Expensive components added to the trouble. Getting a fair deal from mould providers and coming up with mould designs was also not easy.

Time to talk science

Betty is a GSM-enabled 16-ampere monitoring plug-cum-security device, which, when connected to any appliance, helps control it remotely through Be-Connected mobile application or the web application (supporting Android as well as iOS) developed by the company. The solution also provides customers with major consumption data and insights that help them reduce energy wastage, slashing their utility bills and giving them full device control. Betty connects to an input plug-point like any normal plug. The appliance to be used is connected thereafter.

When the user logs into Be-Connected mobile application and sets a schedule, the application connects to the cloud server; the company uses Amazon Web Services. The server triggers a command to Betty, which acts accordingly.

The Betty circuit primarily consists of an SMPS, a GSM-PCB (comprising majorly a GSM modem, current sensors, motion sensor and a microcontroller) and, finally, relay sections. When a device is plugged into Betty, the SMPS scales the current in accordance with the device capacity and forwards it to the GSM-PCB. Current sensors in the PCB detect the current consumed and the information is relayed to

Betty—The energy monitoring and security plug

